

Energy-Aware Adaptive Self-Speculative Decoding

Project summary for understanding and presentation

Motivation:

- Reduce energy use and latency for small language models on laptops and consumer GPUs.
- Keep output quality stable while avoiding wasted speculative work.

Research idea:

- Use early-layer draft generation and full-model verification.
- Choose between ordinary decoding and speculative decoding based on energy cost.
- Use recent acceptance history and context length to decide whether a draft is worth it.

Main research question:

- When does speculative decoding actually save energy per committed token?
- When is ordinary decoding safer and cheaper?

Current prototype status:

- The repo contains a synthetic controller prototype that chooses draft depth and draft length.
- It simulates energy, latency, and acceptance behavior.
- The prototype passes tests and demonstrates the controller logic.
- The real model version is prepared as the next step.

What the project does and how it works

1. Offline preparation

- Profile action candidates such as ordinary decoding, draft depth 4, and draft length 2 or 5.
- Measure acceptance rate, energy, latency, and memory for each configuration.
- Use calibration prompts to fit the controller policy.

2. Online inference

- At each generation step, the controller checks the current context and acceptance history.
- It compares estimated energy per committed token against ordinary decoding.
- If expected gains are uncertain, it falls back to ordinary decoding.

3. Hardware support

- The project is designed for CPU, CUDA, Apple MPS, Intel XPU, and simulation mode.
- An RTX 3050 setup script is included for NVIDIA users.
- Real model execution requires PyTorch, Transformers, and CUDA-enabled tooling.

4. Output and interpretation

- Energy per token, latency, throughput, acceptance rate, action trace, and output matching are reported.
- A good result is lower energy per committed token without changing ordinary output behavior.

Summary:

This project is a focused research prototype for an energy-aware speculative decoding controller, starting from a working synthetic simulator and moving toward a real GPU-enabled evaluation pipeline.